

Self-Healing Infrastructure with Prometheus, Alertmanager & Ansible

Introduction

In modern DevOps environments, maintaining high availability and reducing downtime are critical objectives. This project demonstrates a self-healing infrastructure that automatically detects and recovers from service failures using Prometheus, Alertmanager, and Ansible. The aim is to automate the process of fault detection and recovery, thereby enhancing system reliability and minimizing manual intervention.

Abstract

The Self-Healing Infrastructure project integrates monitoring and automation tools to achieve autonomous recovery. Prometheus monitors system metrics and application uptime, Alertmanager handles alert notifications, and Ansible executes playbooks that perform recovery actions such as restarting services or servers. The workflow ensures that when a service failure or performance degradation occurs, the system self-corrects without human input.

Tools Used

- Prometheus – For real-time monitoring of services and metrics.
- Alertmanager – For managing and triggering alerts based on Prometheus rules.
- Ansible – For executing automated recovery playbooks.
- Shell Scripting – To integrate alert webhooks and automate triggers.
- Docker / Ubuntu VM – For containerized or virtualized environment setup.

Steps Involved in Building the Project

- Deployed a sample NGINX web service on Ubuntu or Docker.
- Installed and configured Prometheus to monitor service uptime and system metrics.
- Created Prometheus alert rules (e.g., service down or CPU usage above 90%).
- Configured Alertmanager to receive alerts and trigger webhook notifications.
- Integrated Alertmanager webhook with an Ansible playbook that restarts the failed service.
- Tested the setup by manually stopping the service and observing automated recovery logs.

Conclusion

The Self-Healing Infrastructure project showcases the power of automation and monitoring tools in maintaining system stability. By integrating Prometheus, Alertmanager, and Ansible, the infrastructure can detect failures and recover autonomously, reducing downtime and improving operational efficiency. This approach forms a foundational concept in Site Reliability Engineering (SRE).